The §4 gate bound: the objects, the counterexamples, and exactly why each proof route fails — worked computations

(secretary worked-reference; all values machine-computed from the faithful model)

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Notation. Characters a, b, c; r^* = star; $r \cdot s$ = sequence; $r + s$ = alternation; $\mathbf{1}$ = the empty-string regex. $|\cdot|$ is regex size; $\|X\| = \sum_{x \in X} |x|$ over the *deduplicated* set X. All numbers below are computed by the validated Python model (`scratch_ce_trace_for_doc.py`).

1 The one inequality we must prove

The D law, the static cubic front, and the gate wrapper are all green. The §1 gate reduces to the single numeric inequality

$$\boxed{\|\text{scd}\ p\ k\| \ \leq\ \text{ctx\_bound}(\text{drain\_ctxs}\ p)\ k} \qquad \text{(TRUE: 0 violations over} > 10^5 \text{ depth-} \geq 5 \text{ cases).}$$

Everything below is about *proving* this true inequality. Four/eight routes failed; this note shows, for each, the exact left/right sides and their values on the counterexamples.

2 The objects (concrete)

S **(`rsimpStrong_raw`).** The strong normaliser. Crucially it *collapses* $a^* \cdot a^* \to a^*$ (a guarded rule, only when the two starred bodies are syntactically equal).

$\diamond_4$ **(`rsimp4_SEQ_atom`).** The *weak* sequence plug: $h \diamond_4 k$ appends k after head h (reassociating), but **never** collapses $a^* \cdot a^*$. (`rsimp7` is the same but *does* collapse, at the top head only.)

drain_ctxs p**.** A *list* of slots $(head_i, cost_i)$, one per "drain weight" unit of p (one per character, one per star), built structurally. $slot\_cost(p, k, i) = cost_i + (1 + |k|)$ and $\text{ctx\_bound}(\text{drain\_ctxs} p)\, k = \sum_i slot\_cost(p, k, i)$.

$\boldsymbol{weak\_slot\_rows}(p, k, i) = \text{rows}(\ \boldsymbol{head_i} \diamond_4 k\ ) \setminus \text{rows}(k)$**.** The *uncollapsed* opening of slot i. wcd $p\ k = \bigcup_i weak\_slot\_rows(p, k, i$
Green fact: $\|\text{wcd}\ p\ k\| \leq \text{ctx\_bound}(\text{drain\_ctxs} p)\, k$.

scd $p\ k = \text{SOL}(\boldsymbol{nseq}\, p\ k) \setminus \text{SOL}(\mathsf{S}k)$**.** The **actual** object: the opened rows after the *strong*, S-normalised step. *These rows are collapsed.*

The whole tension in one line: the weak rows use $\diamond_4$ (uncollapsed), the strong rows use S (collapsed). So a strong row and "its" weak row are often *different rows* of *different sizes*, even though the budget covers them.

3 Three counterexamples, fully computed

All three use $k = a^*$, and all are in-regime ($\mathsf{S}p = p$, $\mathsf{S}k = k$, normal form). For each p the slot ledger drain_ctxs$p$ and ctx_bound are shown, then scd and wcd.

CE-A: $p = b \cdot a^*$ — the collapse, in isolation

$$\text{scd}(b \cdot a^*,\, a^*) = \{\, \underbrace{b \cdot a^*}_{||=4} \,\}, \qquad \|\text{scd}\| = 4.$$

Slots of drain_ctxs$(b \cdot a^*)$ (head, cost, slot_cost, weak row):

$$\begin{array}{llll}
\text{slot}_0: & b \cdot a^*\ (4) & slot_cost = 7 & \text{weak} = b \cdot (a^* \cdot a^*)\ \ || = 7 \\
\text{slot}_1: & a^*\ (2) & slot_cost = 5 & \text{weak} = a^* \cdot a^*\ \ || = 5 \\
\text{slot}_2: & a \cdot a^*\ (4) & slot_cost = 7 & \text{weak} = a \cdot (a^* \cdot a^*)\ \ || = 7
\end{array}$$

ctx_bound $= 7+5+7 = 19$, and wcd $= \{b\cdot(a^*\cdot a^*),\ a^*\cdot a^*,\ a\cdot(a^*\cdot a^*)\}$, $\|\text{wcd}\| = 19$. **Key:** the one strong row $b\cdot a^*$ is the S-*collapse* of the slot$_0$ weak row:

$$\mathsf{S}\big(b\cdot(a^*\cdot a^*)\big) = b\cdot a^*, \qquad |b\cdot a^*| = 4 \ \le\ 7 = |b\cdot(a^*\cdot a^*)|.$$

So $b\cdot a^* \notin$ wcd (set-containment into the weak cover *fails*), but it is the collapse of a weak row, and collapse *shrinks*. Bound holds: $4 \le 19$.

CE-B: $p = b\cdot a^* + \mathbf{1}$ — the RALTS wrap *retains* the uncollapsed row

Same slots/ctx_bound = 19 as CE-A (the $\mathbf{1}$ adds no slot). But now

$$\text{scd}(b\cdot a^* + \mathbf{1},\, a^*) = \{\underbrace{b\cdot(a^*\cdot a^*)}_{||=7}\}, \qquad \|\text{scd}\| = 7.$$

Same head $b\cdot a^*$, opposite outcome: wrapped under $+$, the opener exposes the *uncollapsed* $b\cdot(a^*\cdot a^*)$ and the strong step does *not* collapse it back. And S runs the *wrong way* for any witness:

$$\mathsf{S}\big(b\cdot(a^*\cdot a^*)\big) = b\cdot a^* \ \ne\ b\cdot(a^*\cdot a^*).$$

So *no* weak row y has $\mathsf{S}(y) =$ the strong row (every S-image is the *shorter* $b\cdot a^*$). The strong row here *is* the slot$_0$ weak row itself ($x = y$). Bound holds: $7 \le 19$.

CE-Hall: $p = a + (b^*\cdot a^*)$ — both versions at once

$$\text{scd}(a + b^*\cdot a^*,\, a^*) = \{\underbrace{a\cdot a^*}_{4},\ \underbrace{b^*\cdot a^*}_{5},\ \underbrace{b^*\cdot(a^*\cdot a^*)}_{8}\}, \quad \|\text{scd}\| = 17, \quad \text{ctx\_bound} = 34.$$

The slot for head $b^*\cdot a^*$ is slot$_1$ (cost 5, slot_cost 8, weak row $b^*\cdot(a^*\cdot a^*)$, size 8). Now:

$$b^*\cdot(a^*\cdot a^*) \text{ is the slot}_1 \text{ weak row } (x{=}y), \qquad \mathsf{S}\big(b^*\cdot(a^*\cdot a^*)\big) = b^*\cdot a^*.$$

So *both* strong rows $b^*\cdot a^*$ (collapsed, 5) and $b^*\cdot(a^*\cdot a^*)$ (uncollapsed, 8) have their *only* origin at slot$_1$, whose single budget 8 cannot pay both ($5+8 = 13 > 8$). Bound still holds ($17 \le 34$): the *total* budget covers them because other slots are underused.

4 Each failed route: left side, right side, value on the CE

R1. Per-child set-containment (`master_cover`, the original C-DRAIN). Tried: $\text{scd}(\sum_q q)\,k \subseteq \bigcup_q \text{scd}\ q\ k$ (parent rows are union of child rows). *Intuition:* a $\sum$ should just be the union of its parts. *Killed by CE-B:* LHS $= \{b\cdot(a^*\cdot a^*)\}$ (parent, uncollapsed), but the child $b\cdot a^*$ alone gives $\text{scd}(b\cdot a^*, a^*) = \{b\cdot a^*\}$ (CE-A, collapsed) and $\mathbf{1}$ gives $\{\}$, so RHS $= \{b\cdot a^*\}$. The parent's uncollapsed row $b\cdot(a^*\cdot a^*) \notin \{b\cdot a^*\}$. The $+$-wrap blocks the collapse the child performs. Fails.

R2. Total-size domination. Tried: $\|\text{scd}\ p\ k\| \ \le\ \|\text{wcd}\ p\ k\|$ (then the green $\|\text{wcd}\| \le$ ctx_bound finishes). *Intuition:* strong rows are "smaller" than the weak cover, so their total is too. *False:* S can *grow* a row's opened ledger, so the strong total can exceed the weak total. Machine CE: $p = (\,\mathbf{1} + b\cdot(a+\mathbf{1}) + (\mathbf{1}+a)\cdot a^*\,)\cdot a^*,\ k = c$: $\|\text{scd}\| = 36 \ >\ 29 = \|\text{wcd}\|$ (while ctx_bound $= 51$ still covers 36). So the weak *set* does not dominate; only the *ledger* over-covers.

R3. Injective slot-origin via `collapses_to` (verdict6). Tried: an injection $x \mapsto$ (slot i_x, witness y_x) with $\mathsf{S}(y_x) = \{x\}$ and $|x| \le |y_x|$. *Intuition:* each strong row is the collapse of a paid weak slot row. *Killed by CE-B:* the strong row $x = b\cdot(a^*\cdot a^*)$ is *uncollapsed*; no weak y has $\mathsf{S}(y) = x$ (S only *produces* the shorter $b\cdot a^*$). The relation points the wrong way: x is the long predecessor, not the collapse. (97/40849 failures.)

R4. Two-route injective charge (secretary salvage). Tried: same injection but $x = y_x \lor x \in \text{rows}(\mathsf{S}y_x)$ (allow "x *is* the weak row"). *Intuition:* cover both the uncollapsed ($x{=}y$) and collapsed ($x = \mathsf{S}y$) cases. *Killed by CE-Hall:* $b^*\cdot a^*$ (via $x \in \mathsf{S}y$) and $b^*\cdot(a^*\cdot a^*)$ (via $x{=}y$) are *both* edged only to slot$_1$. Two rows, one slot $\Rightarrow$ no injection (a Hall collision), even though slot$_1$'s cost would suffice for either alone. (24/52851; cuts R3 by $4\times$ but cannot reach 0.)

R5. Pure numeric induction, RALTS step. Tried: $\|\text{scd}(\sum_q q)\,k\| \ \le\ \sum_q \|\text{scd}\ q\ k\|$ (deduped-union size bounded by the sum of children). *Killed by CE-B:* LHS $= \|\{b\cdot(a^*\cdot a^*)\}\| = 7$, RHS $= \|\{b\cdot a^*\}\| + 0 = 4$; $7 > 4$. The parent's single uncollapsed row is *bigger* than the whole child sum. (RALTS has zero constructor slack, so there is no room for the extra.)

R6. Any slot-derived carrier (the impossibility square). For *any* carrier W' built from the slot rows: plain weak $W' = \text{wcd}$ has $\|\text{scd}\| > \|W'\|$ sometimes (R2); S-applied carriers shrink *below* scd (bridge fails worse); wcd ∪ scd pushes the *ceiling* above ctx_bound (CE-A-like: $23 > 19$). No carrier sits between $\|\text{scd}\|$ and ctx_bound. The whole carrier/membership class is dead.

5 The one obstruction, stated once

scd p k is a *heterogeneous* set: a $\sum$/star wrap can make it *retain the uncollapsed long row* $b \cdot (a^* \cdot a^*)$ (CE-B), while a bare sequence makes it *keep the collapsed short row* $b \cdot a^*$ (CE-A), and a mix keeps *both* (CE-Hall). The weak cover only ever holds the uncollapsed forms. Therefore:

- set-containment fails (collapsed strong rows are not weak members);
- S-witnessing fails (uncollapsed strong rows are not S-images);
- any row→slot *injection* fails (collapsed + uncollapsed siblings collide on one slot);
- yet the *total* ledger ctx_bound always covers $\|\text{scd}\|$ (other slots are underused).

Consequence (the search-narrowing result): the proof cannot be a per-row or per-slot *set/membership* statement. It must be a *total-budget* argument that does not match rows to slots.

6 Where verdict7 points

verdict7's "budgeted collapse trace" is exactly this: charge at the level of *syntax-node occurrences → cost tokens* (atoms of ctx_bound), not rows → slots. On CE-Hall the short row $b^* \cdot a^*$ (5 nodes) and the long row $b^* \cdot (a^* \cdot a^*)$ (8 nodes) consume *disjoint* cost atoms (drawn from slot_1 *and* the source-star/collapse slot) instead of both demanding the one indivisible slot_1 — so the Hall collision dissolves. The open question is whether the node→atom injection is constructible at RALTS; that is what to validate before executing.